1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-like lymphoid aggregates in bladder tumors are generally associated with improved local tumor control, but progression and nodal metastasis occur when tumor cells acquire an epigenetic program (high SDC2 promoter methylation leading to SDC2 silencing) that, in concert with elevated GREM1-driven BMP signaling, remodels the extracellular matrix and stromal/vasculature niches to create an immune-refractory microenvironment. This SDC2 methylation–GREM1 axis promotes ECM remodeling, chemokine/cytokine rewiring, and altered BMP pathway activity that dampens TLS effector cell infiltration and/or function (e.g., cytotoxic T cells, NK cells) in the tumor core and peritumoral regions, enabling metastatic escape despite TLS presence. Clinically, TLS-rich bladder cancers with high SDC2 promoter methylation and high GREM1 expression will have shorter progression-free intervals (PFI) and higher nodal involvement than TLS-rich tumors lacking this axis. This effect is modulated by systemic factors (HDL-C and metabolic state) that influence immune cell metabolism and stromal remodeling, and by histologic features (pZVR and SARIFA) that reflect vasculature and adipose–stroma interactions that facilitate dissemination. Urine DNA methylation of SDC2 will correlate with tumor SDC2 methylation and serve as a non-invasive readout of the axis, reinforcing its prognostic and potentially therapeutic relevance. The hypothesis implies testable predictions: (1) in TLS-high tumors, SDC2 methylation and GREM1 expression co-segregate with worse PFI; (2) restoring SDC2 expression or inhibiting GREM1/BMP signaling will restore TLS efficacy and reduce metastatic spread in model systems; (3) systemic metabolic state and HDL-C levels will modify the strength of the TLS–SDC2/GREM1 interaction; (4) urine SDC2 methylation status tracks tissue methylation and correlates with aggressive biology.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	GREM1, SDC2
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GREM1: GREM1 (BMP antagonist linked to nodal metastasis) — offers a pro-metastatic, microenvironment-remodeling axis that can impair immune efficacy.

SDC2: SDC2/TFPI2 methylation (epigenetic tumor signatures) — provides a mechanism for tumor-intrinsic programs that could confer resistance to immune control; (c) GREM1 (BMP antagonist linked to nodal metastasis) — offers a pro-metastatic, microenvironment-remodeling axis that can impair immune efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	selected urine-DNA methylation biomarkers	Supporting modifier from knowledge graph
Secondary	SARIFA-positivity	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TFPI2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.